

EXPERIMENT - 22

1. Problem Statement

Write a menu-driven C program that accepts two positive integers, m and n , where $m \leq n$. Display the following menu and use a switch statement to perform the operation selected by the user:

- Print the multiplication tables of m and n side by side, using multipliers from 1 to 10.
- Display a multiplication grid containing the products of integers from 1 to m and integers from 1 to n .
- For each integer from m to n , print that integer followed by the next five consecutive integers.
- Exit the program.

Continue displaying the menu until the user selects the Exit option. Display an appropriate error message if the user enters an invalid menu choice.

2. Algorithm

Step 1: START

Step 2: Read m .

Step 3: While m is less than or equal to 0, display the error message and read m again.

Step 4: Read n .

Step 5: While n is less than or equal to 0, or n is less than m , display the matching error message and read n again.

Step 6: Repeat Steps 7 to 9 using a do-while loop, until the choice is 4.

Step 7: Display the menu and read choice:

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Step 8: Using switch (choice), do the following:

case 1: For i from 1 to 10, print $m \times i$ and $n \times i$ in the same line, so that the two tables come side by side.

case 2: First print the column headings 1 to n and a line below them. Then for i from 1 to m , print i as the row heading and inside it, for j from 1 to n , print $i \times j$. This gives a grid of m rows and n columns.

case 3: For i from m to n , print i , and inside it, for j from 0 to 5, print $i + j$. This prints the integer followed by the next five consecutive integers, that is six numbers in every line.

case 4: Display the exit message.

default: Display "Invalid choice! Enter 1 to 4."

Step 9: Display the result of the chosen operation.

Step 10: STOP

3. Test Case Table

Valid inputs

TC #	m and n	Choice	Expected Output	Actual Output	Result
TC1	m = 3, n = 5	1	Table of 3 on the left (3 to 30) and table of 5 on the right (5 to 50)	3 x 1 = 3 5 x 1 = 5 up to 3 x 10 = 30 5 x 10 = 50	PASS
TC2	m = 3, n = 5	2	A grid of 3 rows and 5 columns, last row 3, 6, 9, 12, 15	Row 1: 1 2 3 4 5 Row 2: 2 4 6 8 10 Row 3: 3 6 9 12 15	PASS
TC3	m = 3, n = 5	3	3: 3 4 5 6 7 8 4: 4 5 6 7 8 9 5: 5 6 7 8 9 10	3: 3 4 5 6 7 8 4: 4 5 6 7 8 9 5: 5 6 7 8 9 10	PASS
TC4	m = 3, n = 5	4	Exiting the program.	Exiting the program.	PASS

Boundary inputs

TC #	m and n	Choice	Expected Output	Actual Output	Result
TC5	m = 4, n = 4 (m is equal to n)	2	A square grid of 4 rows and 4 columns	Last row : 4 8 12 16	PASS
TC6	m = 4, n = 4	3	Only one line, because there is only one integer from m to n	4: 4 5 6 7 8 9	PASS
TC7	m = 1, n = 1 (the smallest allowed values)	2	A grid of only one box	1 1	PASS
TC8	m = 1, n = 1	3	1: 1 2 3 4 5 6	1: 1 2 3 4 5 6	PASS

Invalid inputs

In every case below the program shows the error message and asks for the same value again. It does not stop and it does not take the wrong value.

TC #	Invalid input given	Expected Output	Actual Output	Result
TC9	m = -5, then 3	Error message, then ask again	Error: m must be a positive integer. Enter again:	PASS
TC10	m = -5, then 0, then 3 (invalid two times)	Error message both the times, then accept 3	The error message was shown 2 times, then it accepted 3	PASS
TC11	m = 3, n = 2 (n is smaller than m)	Error message, then ask again	Error: n must not be less than m (3). Enter again:	PASS
TC12	m = 3, n = -1, then 5	Error message, then ask again	Error: n must be a positive integer. Enter again:	PASS
TC13	Menu choice = 9	Invalid choice message, and the menu is shown again	Invalid choice! Enter 1 to 4.	PASS

4. Source Code

grid_tables.c (93 lines)

```
/* Menu-driven program that accepts two positive integers m and n,
   where m is not greater than n, and prints the multiplication
   tables of m and n side by side, a multiplication grid, and the
   consecutive integers from m to n. */

#include <stdio.h>

int main(void)
{
    int m, n, choice, i, j;

    /* read and validate m */
    printf("Enter the value of m: ");
    scanf("%d", &m);
    while (m <= 0) {
        printf("Error: m must be a positive integer. Enter again: ");
        scanf("%d", &m);
    }

    /* read and validate n, it must be positive and not less than m */
    printf("Enter the value of n: ");
    scanf("%d", &n);
    while (n <= 0 || n < m) {
        if (n <= 0)
            printf("Error: n must be a positive integer. Enter again: ");
        else
            printf("Error: n must not be less than m (%d). Enter again: ", m);
        scanf("%d", &n);
    }

    do {
        printf("\n1. Multiplication tables of m and n\n");
        printf("2. Multiplication grid\n");
        printf("3. Consecutive integers\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {

            case 1:
                printf("Multiplication Tables of %d and %d (m=%d, n=%d)\n",
                    m, n, m, n);
                for (i = 1; i <= 10; i++)
                    printf("%d x %-2d = %-8d %d x %-2d = %d\n",
                        m, i, m * i, n, i, n * i);
                break;

            case 2:
                printf("Multiplication Grid (m=%d, n=%d)\n", m, n);

                /* the column headings */
                printf("    ");
                for (j = 1; j <= n; j++)
                    printf("%5d", j);
                printf("\n    ");
                for (j = 1; j <= n; j++)
                    printf("-----");
                printf("\n");

                /* one row for every integer from 1 to m */
                for (i = 1; i <= m; i++) {
                    printf("%3d|", i);
                    for (j = 1; j <= n; j++)
                        printf("%5d", i * j);
                    printf("\n");
                }

            }
    } while (choice != 4);
}
```

```
        break;

    case 3:
        printf("Consecutive Integers from %d to %d (m=%d, n=%d)\n",
            m, n, m, n);
        for (i = m; i <= n; i++) {
            printf("%d: ", i);
            for (j = 0; j <= 5; j++)
                printf("%d ", i + j);
            printf("\n");
        }
        break;

    case 4:
        printf("Exiting the program.\n");
        break;

    default:
        printf("Invalid choice! Enter 1 to 4.\n");
        break;
}

} while (choice != 4);

return 0;
}
```

5. Compilation and Output

I wrote the program in the `MyAssignment` folder using the `nano` editor, then compiled and ran it:

```
$ cd ~/Desktop/MyAssignment
$ nano grid_tables.c
$ gcc grid_tables.c -o grid_tables
$ ./grid_tables
```

The program compiled in the first attempt itself, without any error or warning.

All the three options, with $m = 3$ and $n = 5$

```
MyAssignment — -zsh — 122x31

adilshakil@ADILs-MacBook-Air MyAssignment % ./grid_tables
Enter the value of m: 3
Enter the value of n: 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1
Multiplication Tables of 3 and 5 (m=3, n=5)
3 x 1 = 3      5 x 1 = 5
3 x 2 = 6      5 x 2 = 10
3 x 3 = 9      5 x 3 = 15
3 x 4 = 12     5 x 4 = 20
3 x 5 = 15     5 x 5 = 25
3 x 6 = 18     5 x 6 = 30
3 x 7 = 21     5 x 7 = 35
3 x 8 = 24     5 x 8 = 40
3 x 9 = 27     5 x 9 = 45
3 x 10 = 30    5 x 10 = 50

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2
Multiplication Grid (m=3, n=5)
      1   2   3   4   5
-----
1|    1   2   3   4   5
2|    2   4   6   8  10
3|    3   6   9  12  15

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3
Consecutive Integers from 3 to 5 (m=3, n=5)
3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 4
Exiting the program.
adilshakil@ADILs-MacBook-Air MyAssignment %
```

Boundary cases

```
MyAssignment — -zsh — 122x31

adilshakil@ADILs-MacBook-Air MyAssignment % ./grid_tables
Enter the value of m: 4
Enter the value of n: 4
Enter your choice: 2
Multiplication Grid (m=4, n=4)
  1   2   3   4
-----
1|  1   2   3   4
2|  2   4   6   8
3|  3   6   9  12
4|  4   8  12  16

Enter your choice: 3
Consecutive Integers from 4 to 4 (m=4, n=4)
4: 4 5 6 7 8 9

Enter your choice: 4
Exiting the program.
adilshakil@ADILs-MacBook-Air MyAssignment % ./grid_tables
Enter the value of m: 1
Enter the value of n: 1
Enter your choice: 2
Multiplication Grid (m=1, n=1)
  1
----
1|  1

Enter your choice: 3
Consecutive Integers from 1 to 1 (m=1, n=1)
1: 1 2 3 4 5 6

Enter your choice: 4
Exiting the program.
adilshakil@ADILs-MacBook-Air MyAssignment %
```

The menu is displayed again before every choice. It is shown fully only in the first run above, so that the output stays readable.

Invalid inputs

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./grid_tables
Enter the value of m: -5
Error: m must be a positive integer. Enter again: 0
Error: m must be a positive integer. Enter again: 3
Enter the value of n: 2
Error: n must not be less than m (3). Enter again: -1
Error: n must be a positive integer. Enter again: 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 9
Invalid choice! Enter 1 to 4.

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3
Consecutive Integers from 3 to 5 (m=3, n=5)
3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10

Enter your choice: 4
Exiting the program.
adilshakil@ADILs-MacBook-Air MyAssignment %
```